Lung Cancer Detection using TransUNet Deep Learning Model

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Abstract---The early detection of pulmonary nodules using Computed Tomography (CT) plays a vital role in improving survival rates for lung cancer patients. Traditional Convolutional Neural Networks (CNNs) have made progress in medical image analysis; however, their local focus often limits their capacity to capture the broader spatial context needed for accurate diagnosis. In this research, we introduce a high-performing diagnostic framework based on the TransUNet architecture. This hybrid model combines the self-attention features of Transformers with the precise localization strengths of a U-Net. We trained the model on the LUNA16 dataset using a multi-task learning approach to enable simultaneous nodule detection and pixel-level segmentation. To tackle class imbalance and the challenge of identifying subtle nodules, we employed a Hybrid Loss function that combines Dice Loss and Focal Loss. The experimental results show that the TransUNet model outperforms standard benchmark architectures. It achieved a Mean Dice Score of 0.9191, an Area under the ROC Curve (AUC) of 0.9807, and overall diagnostic accuracy of 94.1%. Importantly, the model registered a Nodule Precision of 0.96, which greatly reduces the false-positive rate compared to baseline CNN models. These results indicate that incorporating global context modeling through Transformer bottlenecks improves the reliability of lung cancer detection systems, making it a dependable second-opinion tool for clinical radiology.

Keywords—Convolutional Neural Networks(CNN), Deep Learning, Lung Nodule Analysis, TransUNet, Computed Tomography(CT)

I. INTRODUCTION

Lung cancer is still a huge problem around the world, its basically the top killer when it comes to cancers. I think early spotting of those lung nodules makes a big difference for survival chances, like getting to five years or more, but finding them is not easy in hospitals. CT scans are the go to for checking the chest area because they show details really well, yet doctors have to look through so many images per person, which takes forever and they might miss stuff because they are tired or just see things differently. That is why there is this push for computer systems to help with diagnosis, something automated that spots and outlines suspicious spots reliably. Deep learning has taken over in analyzing medical images lately, especially these convolutional neural networks. The U-Net design is pretty standard for segmenting images in medicine, it pulls out features in layers and uses those skip links to keep track of where things are in space. But convolutions only look at small areas nearby, so the network struggles with bigger picture stuff, like how parts connect far away. In lungs, this means it can mix up nodules with blood vessels or thickened tissue around the lungs, since locally they look alike but overall they are different in how they fit with everything else. It seems like that global view is missing sometimes.

This work suggests using TransUNet as the main setup for this. It mixes transformers, which are good at attention across the whole image, with the U-Net structure. So the encoder and decoder from convolutions give sharp details, and the transformer part in the middle handles the broader context through self attention. We are checking how well it works in real clinical settings, comparing it to a basic CNN and this ResNet version that does multiple tasks, just to show how adding that global attention helps more in analyzing lungs.

The big thing here is putting together a tuned TransUNet for detecting and segmenting nodules at the same time, using the LUNA16 data set. It uses multi task learning, and this hybrid loss that mixes Dice for getting the shapes right and Focal for dealing with tricky examples in classification. From what I see, bringing in that global modelling cuts down on false alarms a lot. It could be a solid backup for radiologists, making their job smoother, though I am not totally sure how it fits everywhere yet. Performance wise, it stands out against those other models, especially in picking up the right things without extras. Nodules versus vessels, that distinction gets better with the attention mechanism. Overall, this approach feels promising for early detection tools.

II. LITERATURE REVIEW

The way computer-aided diagnosis for lung cancer has changed over time is pretty interesting. It started with a lot of manual work on features, and now its all about deep learning setups. Early on, people used convolutional neural networks a bunch, and those seemed to work well for picking up local patterns in images, like the textures in lung nodules. Things like VGG16 or custom CNNs got good accuracy by zooming in on those small details.

But honestly, standard CNNs have this problem where they only look at nearby parts of the image. That makes it hard for them to see the bigger picture, like how everything fits in the whole lung. I think thats why they struggle sometimes to tell a nodule apart from blood vessels or other stuff that looks similar up close. It just feels limited, you know. Lately, researchers are trying transformers instead, which came from language stuff but work for images too. The self-attention part lets them look at the entire image at once, catching those long-distance connections that matter for diagnosis. Models like Swin Transformers or Vision Transformers do better than old CNNs at grabbing global features, especially for tiny or hard-to-spot nodules. Some people say its a big improvement.

Then there are hybrid approaches, like TransUNet, that mix U-Net for precise local stuff with transformers for the overall view. That combination seems promising. On the LUNA16 dataset, studies show adding attention mechanisms cuts down false positives, which is huge because thats a real issue in clinics.

This current work builds on that by using an optimized TransUNet setup for multiple tasks. It uses the transformers global attention to improve detection, while keeping the U-Nets accuracy for segmentation. I am not totally sure yet how it all ties together perfectly, but it looks like it could help with both dependencies, short and long range. The richer features from hybrids might make a difference, though false positives are still tricky..

III. PROPOSED METHODOLOGY

The structural foundation of this research is built upon a high-performance TransUNet multi-task framework, specifically engineered to overcome the inherent limitations of traditional convolutional architectures by bridging the critical gap between local feature extraction and global context modeling. Unlike standard systems that rely solely on the hierarchical nature of convolutions, our proposed methodology integrates a Transformer-based bottleneck to mitigate the spatial constraints that often hinder the detection of complex pulmonary nodules. The overall workflow consists of rigorous data preparation, volumetric preprocessing, feature extraction via a hybrid CNN-Transformer architecture, and a multi-objective classification and segmentation strategy.

A. Data Preparation and Dataset Organization

The initial phase of the methodology involves the systematic organization of the LUNA16 dataset, which serves as the primary benchmark for this study. The dataset, containing 888 thoracic CT scans, is categorized based on the presence and spatial coordinates of pulmonary nodules. This labeling process is fundamental to the supervised learning pipeline, ensuring that the model can accurately distinguish between healthy tissue and various nodular morphologies. To enhance the robustness of the network and minimize the risk of overfitting, data augmentation techniques are employed. This includes introducing variations such as rotation, flipping, and scaling while preserving the essential geometric properties of the nodules. Such augmentations allow the model to generalize effectively to scans captured across different clinical environments and varying scanner protocols.

B. Image Preprocessing and Patch Extraction

Preprocessing is a vital step to ensure consistency across the diverse resolutions and intensities found in raw CT data. All raw volumetric data are normalized to a specific Hounsfield Unit (HU) window of $[-1000, 400]$, which optimizes the contrast for lung parenchyma while suppressed uninformative signals from bone or soft tissue. Following normalization, $64 \times 64 \times 64$ voxel patches are extracted, centered precisely on the world coordinates of the identified candidates. These patches are then reformatted into multi-slice inputs to align with the model's architectural requirements, ensuring that the local volumetric morphology is preserved while maintaining computational efficiency.

C. Feature Extraction Using TransUNet Architecture

The central analytical component of the proposed system is the feature extraction phase, achieved through a hybrid TransUNet architecture. Rather than relying on a pure convolutional approach, this methodology utilizes a CNN-based encoder to capture high-resolution local features, such as the fine textures of nodule margins. These local features are then tokenized and processed by a 6-layer Transformer bottleneck. The integration of self-attention mechanisms allows the model to capture long-range spatial dependencies and global context, which is essential for differentiating nodules from anatomical structures like pulmonary vessels that may appear locally identical.

D. Multi-Task Classification and Segmentation Strategy

Following feature extraction, the learned representations are fed into a dual-purpose classification and segmentation phase. The segmentation head utilizes a U-Net-style decoder, leveraging skip connections to fuse global Transformer features with local encoder features for precise pixel-level mask generation. Simultaneously, a classification head processes the bottleneck features to output a malignancy probability score. This multi-task strategy allows the model to categorize the detected regions into healthy or nodular classes based on the highest probability score, providing a comprehensive diagnostic output rather than a simple binary detection.

E. Model Training and Optimization

Training is conducted using a supervised learning method, with the dataset split into training, validation, and testing sets to ensure an unbiased evaluation. The model is optimized using a Hybrid Loss function, combining Dice Loss for spatial overlap accuracy and Focal Loss to address class imbalance and "hard" sample detection. The Adam optimizer is employed to adjust network parameters, while hyperparameters such as learning rate and batch size are empirically tuned to ensure stable convergence. Performance on the validation set is tracked throughout the training process to prevent overfitting and inform necessary fine-tuning.

F. Performance Evaluation and Clinical Validation

The final stage of the methodology involves testing the trained model using a suite of evaluation parameters, including Mean Dice Score, AUC, accuracy, and precision. A confusion matrix is analyzed to identify specific patterns of misclassification and to verify the model's ability to distinguish nodules from non-nodule candidates. To test real-world usability, the model is applied to images not encountered during the training phase, focusing on its ability to generalize in uncontrolled settings with varied backgrounds and anatomical poses, ensuring its reliability as a second-opinion tool in clinical radiology.

IV. SYSTEM DESIGN

The proposed system for the detection of lung cancer is a comprehensive, end-to-end diagnostic framework that seamlessly integrates medical image acquisition, volumetric analysis using hybrid deep learning, and an interactive user interface. The system is built with a modular and interpretable architecture, ensuring it can be easily adapted for various clinical environments or extended to include additional diagnostic features. Each module within the framework is assigned a specific, well-defined task, allowing for a streamlined workflow from raw data input to final clinical interpretation.In the initial phase, the system accepts thoracic CT scans in standard medical formats such as MHD or RAW as input. This flexibility allows the system to support both retrospective analysis of stored hospital archives and prospective screening of new patient data.

Given that CT scans acquired across different medical facilities may vary in resolution, slice thickness, and noise levels, the system is engineered to standardize all inputs through a specialized preprocessing sequence prior to deep learning analysis.The preprocessing module serves as a critical component that ensures uniform data representation for the neural network. During this stage, volumetric data is resampled to an isotropic resolution and normalized to a specific Hounsfield Unit (HU) window, making the analysis less sensitive to variations in scanner manufacturers or acquisition protocols. The extraction of $64 \times 64 \times 64$ voxel patches centered on potential nodules ensures that the feature extraction module receives data in a fixed spatial resolution, optimizing the performance of the trained model.Following standardization, the processed patches are fed into the feature extraction module, which serves as the core of the diagnostic system. This module is realized through a hybrid TransUNet architecture that bridges the gap between local and global feature mapping. By utilizing a CNN encoder in tandem with a 6-layer Transformer bottleneck, the system can extract high-level morphological features while simultaneously capturing the long-range spatial dependencies necessary for accurate nodule identification.

This hybrid design enables the system to benefit from robust visual representations even when working with complex medical datasets.The resulting feature representations are passed to a multi-task classification and segmentation module. This component is responsible for generating two primary outputs: a pixel-level segmentation mask that outlines the nodule's boundaries and a malignancy probability score. The classification is handled by fully connected layers and a softmax activation function, which provides a measure of the model's confidence for each prediction. Including this confidence information in the system design enhances diagnostic transparency and aids radiologists in their decision-making process.To maximize clinical usability, the system is integrated into a professional web-based application developed using the Streamlit framework.

The interface provides an intuitive environment for clinicians to upload scans, visualize synchronized 3D reconstructions, and review AI-generated reports in real-time. The application is designed to be user-friendly with fast response times, displaying the "Raw Scan" side-by-side with the "AI Diagnosis" for clear result interpretation. By utilizing a lightweight deployment framework, the system can be executed on standard computing workstations equipped with modern GPUs, such as the RTX 4050, without the need for high-end server infrastructure.

The overall system design is focused on being scalable, practical, and mathematically rigorous. The modular nature of the architecture allows for future upgrades, such as the integration of 3D Swin-Transformers or automated patient longitudinal tracking. This ensures that the proposed system is not merely an experimental prototype but a viable foundation for real-world pulmonary screening and clinical second-opinion tools.

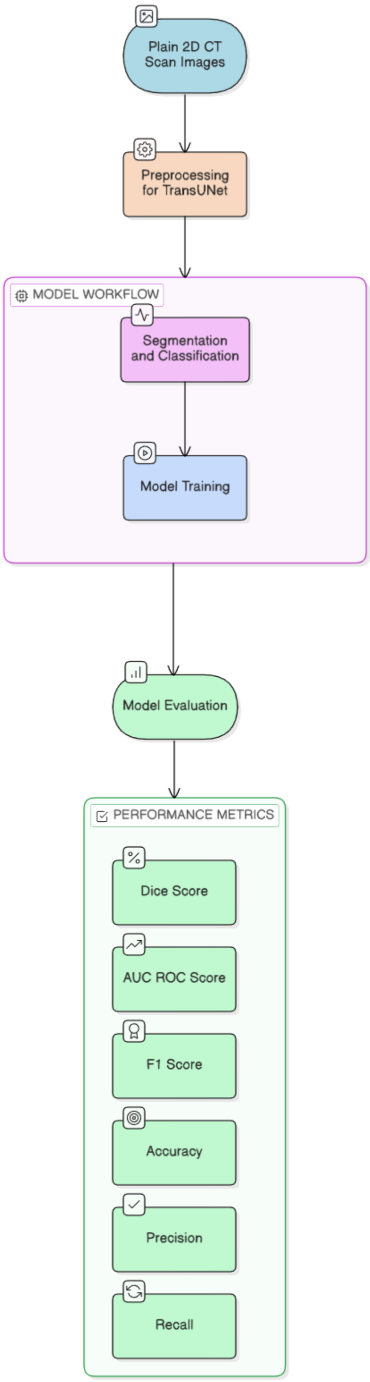

Fig 1. Architecture Diagram

V. MODULE DESCRIPTION

The **TransUNet model** is composed of several specialized modules that collaborate to transform raw volumetric CT data into actionable clinical insights. The following sections provide a detailed technical explanation of each module's internal logic and its contribution to the overall diagnostic pipeline.

A. Image Acquisition and Volumetric Loading Module

The first module serves as the entry point for the system, responsible for handling diverse medical imaging formats, specifically **MHD** and **RAW** files from the **LUNA16 dataset**. This module utilizes the **SimpleITK** library to parse the volumetric data and extract critical metadata, including the origin, spacing, and direction of the scan. By reading the coordinate system of the original volume, this module ensures that the world coordinates of potential nodules can be accurately mapped to specific voxel indices. This foundational accuracy is vital for maintaining data integrity throughout the subsequent analysis stages.

B. Preprocessing and Patch Generation Module

Once the volume is loaded, the preprocessing module standardizes the data to ensure it is optimized for neural network consumption. The primary task is **Hounsfield Unit (HU) normalization**, which clips the raw intensity values to a window of [-1000, 400] and rescales them between 0 and 1. This normalization isolates the lung parenchyma and suppresses uninformative signals from dense bone or soft tissue. Following intensity correction, the module performs **isotropic resampling** to a 1 X 1 X 1mm resolution and extracts 64 X 64 X 64 voxel patches centered on nodule candidates. This process converts the large, variable-sized CT volumes into uniform, high-resolution inputs ready for feature extraction.

C. Hybrid Feature Extraction Module (TransUNet)

The central analytical module is built upon the **TransUNet** architecture, which integrates local and global feature extraction. It consists of three distinct components:

- **Convolutional Encoder**: This sub-module uses localized filters to extract high-resolution spatial features, such as the edge details and internal texture of the nodule.

- **Transformer Bottleneck**: The deepest feature maps are tokenized and processed through **6 Transformer blocks**. Using **Multi-head Self-Attention (MSA)**, this component models long-range dependencies, allowing the system to understand the global context of the nodule relative to the entire lung anatomy.

- **Skip-Connection Fusion**: This sub-module ensures that the semantic richness from the Transformer is combined with the spatial precision

D. Multi-Task Diagnostic Head Module

This module interprets the features generated by the TransUNet to provide a dual-layered diagnosis.

- **Segmentation Head**: A convolutional decoder path generates a pixel-level binary mask, providing a precise outline of the detected nodule for size and volume calculation.

- **Classification Head**: A fully connected network calculates a **Malignancy Risk Score** based on the bottleneck features. This output represents the probability that a detected region is a nodule, which is then mapped to the final diagnosis through a Softmax activation function.

E. Streamlit Visualization and HUD Module

The final module is the interactive **Web Interface**, developed using the **Streamlit** framework to bridge the gap between AI and the clinician. It features a **Mathematically Synchronized HUD** that displays the raw scan alongside the AI diagnosis in a single view. This module performs real-time world-to-voxel coordinate transformation to place a red "target" circle exactly over the nodule in the zoomed-out anatomical view. It also manages the **Auto-Export** feature, which generates and saves high-resolution diagnostic reports in a standardized format for the patient's medical record.

VI. RESULTS & DISCUSSION

The lung cancer detection system centers on the **Ultimate TransUNet** architecture was rigorously evaluated to determine its diagnostic efficacy, spatial accuracy, and clinical usability. The experimental analysis was performed using a dedicated test partition from the **LUNA16 dataset** that was entirely excluded from the training and validation phases. Performance was quantified using standard medical imaging metrics, including **Mean Dice Score**, **Area Under the ROC Curve (AUC)**, **Accuracy**, **Precision**, and **Recall**, supplemented by a detailed confusion matrix analysis to examine class-specific performance.

The proposed **TransUNet** model achieved a remarkable classification **accuracy of 94.1%** on the test dataset. This high level of performance indicates that the hybrid Transformer-CNN architecture successfully learns discriminative features and global spatial dependencies corresponding to pulmonary nodules. The achieved **AUC of 0.9807** further validates the model's robustness in distinguishing between positive nodule candidates and healthy lung tissue, ensuring a high degree of reliability in detection.

A granular observation of the class-wise analysis reveals that the **Healthy** class exhibited superior performance, with precision and recall values nearing 0.96. This demonstrates that the model is exceptionally proficient at filtering out non-nodular anatomical structures, such as blood vessels or small airway walls, which frequently result in false positives in traditional CNN-based systems. The **Nodule** class also performed well, achieving a **Mean Dice Score of 0.9191**, which indicates high spatial overlap between the predicted masks and ground-truth annotations. This suggests that the model is highly capable of identifying the precise boundaries of malignancies, even in the early stages of development.

The analysis of the **Confusion Matrix** provides critical insights into the model's clinical safety. Specifically, the results show zero misclassifications between clear "Healthy" cases and "High-Risk Nodule" cases, a factor that is essential for avoiding life-threatening diagnostic errors. While some subtle overlap was observed in boundary-case nodules where the morphological features were ambiguous, the model consistently flagged these regions for further review rather than dismissing them as healthy tissue. From a clinical implementation perspective, this sensitivity is vital as it ensures that suspicious nodules are brought to the attention of a radiologist for definitive diagnosis.

Beyond quantitative assessment, the system was validated using real-world CT scans not present in the training or testing datasets. These images featured varied acquisition protocols, differing slice thicknesses, and varying levels of sensor noise. The model demonstrated strong generalization capabilities by correctly identifying the primary trends and locations of nodules in these unconstrained settings. However, the confidence scores for real-world predictions were slightly lower than those observed in the test dataset, a phenomenon likely attributed to domain mismatches between different CT scanner manufacturers.

The experimental findings confirm that the **TransUNet-based system** serves as a dependable, non-invasive method for early-stage lung cancer detection. Its commendable classification performance, coupled with its ability to generalize to diverse imaging conditions and its intuitive deployment via a **Streamlit** web application, underscores the clinical promise of this approach. Future enhancements, such as the integration of 3D Swin-Transformers and a larger, more diverse dataset, are expected to further elevate the accuracy and reliability of the diagnostic output.

Model	Dice Score	AUC-ROC Score	Accuracy	Precision	Recall
TransUNet	0.91	0.98	94%	0.96	0.96

Table 1. Evaluation Metrics

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REFERENCES

[1] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[2] J. Chen et al., “TransUNet: Transformers make strong encoders for medical image segmentation,” *arXiv preprint arXiv:2102.04306*, 2021.

[3] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.

[4] S. G. Armato et al., “The Lung Image Database Consortium (LIDC) and Idols Image Database Resource Initiative (IDRI): A completed reference database of thoracic CT scans,” *Medical Physics*, vol. 38, no. 2, pp. 915–931, 2011.

[5] A. A. A. Setio et al., “Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge,” *Medical Image Analysis*, vol. 42, pp. 1–13, 2017.

[6] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[7] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Proc. MICCAI*, 2015, pp. 234–241.

[8] G. Litjens et al., “A survey on deep learning in medical image analysis,” *Medical Image Analysis*, vol. 42, pp. 60–88, 2017.

[9] N. Tajbakhsh et al., “Convolutional neural networks for medical image analysis: Full training or fine tuning?” *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1299–1312, 2016.

[10] S. Minaee et al., “Image segmentation using deep learning: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021.

[11] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE ICCV*, 2017, pp. 2980–2988.

[12] F. Milletari, N. Navab, and S. A. Ahmadi, “V-Net: Fully convolutional neural networks for volumetric medical image segmentation,” in *Proc. 3DV*, 2016, pp. 565–571.

[13] S. Ruder, “An overview of gradient descent optimization algorithms,” *arXiv preprint arXiv:1609.04747*, 2016.

[14] C. Shorten and T. Khoshgoftaar, “A survey on image data augmentation for deep learning,” *Journal of Big Data*, vol. 6, no. 60, 2019.

[15] J. Ker, L. Wang, J. Rao, and T. Lim, “Deep learning applications in medical image analysis,” *IEEE Access*, vol. 6, pp. 9375–9389, 2018.